

Hybrid Multiobjective Optimization Algorithm and Real life Application

A Thesis submitted by

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In partial fulfilment of the requirements for the degree of

M.Sc - M.Tech



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Declaration

I hereby declare that the Thesis entitled “**A Hybrid Multiobjective Optimization Framework Using NSGA-III and Differential Evolution for Hyperparameter Optimization in Deep Learning**”, submitted in partial fulfilment of the requirements for the award of the degree of **M.Sc-M.Tech** at the **Indian Institute of Technology Jodhpur**, is a record of the original work carried out by me under the supervision of **Dr. Md. Abu Thalamuddin Ansary**, Department of Mathematics.

The work presented in this thesis focuses on the development and analysis of hybrid multi-objective optimization techniques, specifically NSGA-III and its extension using Differential Evolution (DE-NSGA-III), and their application to hyperparameter optimization of deep learning models using the MNIST dataset.

I further declare that this work has not been submitted, either in part or in full, to any other institution or university for the award of any degree, diploma, or certificate. All sources of information used in this report have been duly acknowledged.

The work presented in this dissertation has been carried out by me during the academic session 2025–2026, and I take full responsibility for the content and results presented herein.

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Certificate

This is to certify that the thesis entitled “**A Hybrid Multiobjective Optimization Framework Using NSGA-III and Differential Evolution for Hyperparameter Optimization in Deep Learning**” submitted by **Amit Adhikari (M22MA201)** to the **Indian Institute of Technology Jodhpur**, in partial fulfilment of the requirements for the award of the degree of **M.Sc–M.Tech**, is a bonafide record of the research work carried out by him under my supervision.

The work presented in this thesis focuses on the development and analysis of hybrid multi-objective optimization techniques, specifically the integration of NSGA-III with Differential Evolution (DE-NSGA-III), and their application to hyperparameter optimization of deep learning models. The results demonstrate improved convergence, enhanced exploration, and better performance in terms of accuracy, F1-score, and model complexity.

To the best of my knowledge, the contents of this thesis, in full or in part, have not been submitted to any other Institute or University for the award of any degree or diploma.

Dr. Md. Abu Thalamuddin Ansary

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Abstract

Multiobjective optimization plays a crucial role in solving real-world problems involving multiple conflicting objectives such as accuracy, computational efficiency, and model complexity. In the context of deep learning, hyperparameter tuning is inherently a multiobjective problem, where improving one objective often leads to trade-offs in others.

In this work, we study evolutionary multiobjective optimization techniques, particularly **NSGA-II** and its extension **NSGA-III**, which is designed for many-objective optimization problems using reference-point-based selection. While NSGA-III provides better diversity and well-distributed Pareto-optimal solutions, it suffers from relatively slow convergence due to its reliance on traditional genetic operators such as crossover and mutation.

To address this limitation, we consider a hybrid approach, namely **DE-NSGA-III**, which integrates **Differential Evolution (DE)** with NSGA-III. In this framework, DE-based mutation and selection strategies are incorporated to improve convergence speed and enhance exploration of the search space, while NSGA-III maintains diversity through reference-point-based niching.

The proposed approach is applied to a real-world problem of **multiobjective hyperparameter optimization of a Multi-Layer Perceptron (MLP)** model using the **MNIST dataset**. The objectives include maximizing accuracy and F1-score while minimizing model complexity.

Experimental results demonstrate that the hybrid DE-NSGA-III algorithm achieves:

- Higher accuracy and F1-score
- Faster convergence compared to NSGA-III
- Improved overall performance in terms of weighted objective metrics

These results highlight that hybrid evolutionary approaches provide an effective and scalable solution for complex multiobjective optimization problems in deep learning.

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Chapter 1

Introduction

Optimization plays a crucial role in modern science, engineering, and machine learning, where the goal is to find the best possible solution under given constraints. In many real-world problems, especially in deep learning, multiple performance criteria must be optimized simultaneously. For example, improving classification accuracy may increase model complexity or computational cost. Such problems naturally lead to the framework of *multiobjective optimization*.

Unlike single-objective optimization, where a single optimal solution is obtained, multiobjective optimization aims to find a set of trade-off solutions known as the **Pareto optimal solutions**. These solutions represent different compromises among conflicting objectives such as accuracy, F1-score, and model complexity.

Evolutionary algorithms have emerged as powerful tools for solving multiobjective optimization problems due to their population-based nature. Among them, the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)** is one of the most widely used methods. However, NSGA-II performs effectively only for problems with a small number of objectives and struggles when the number of objectives increases.

To address this limitation, **NSGA-III** was proposed as an extension of NSGA-II. NSGA-III introduces a reference-point-based selection mechanism that ensures better diversity and a well-distributed Pareto front in many-objective optimization problems. Despite its advantages, NSGA-III relies on traditional genetic operators such as crossover and mutation, which may lead to slow convergence.

To improve convergence speed and search efficiency, hybrid approaches have been developed. In this work, we consider a hybrid algorithm known as **DE-NSGA-III**, which integrates

Differential Evolution (DE) with NSGA-III. Differential Evolution enhances the exploration capability of the algorithm by generating new candidate solutions through vector-based mutation and recombination strategies. This allows the algorithm to converge faster while maintaining diversity.

The proposed hybrid framework is applied to the problem of **hyperparameter optimization in deep learning**. Hyperparameter tuning is inherently a multiobjective problem, as it involves balancing multiple competing objectives such as maximizing accuracy and F1-score while minimizing model complexity.

In this work, we use a **Multi-Layer Perceptron (MLP)** model and evaluate performance on the **MNIST dataset**, which is a standard benchmark for handwritten digit classification. The effectiveness of the proposed hybrid approach is demonstrated through improved performance metrics and faster convergence compared to standard NSGA-III.

1.1 Multiobjective Optimization

A general multiobjective optimization problem can be formulated as:

$$\min_{x \in X} F(x) = (f_1(x), f_2(x), \dots, f_m(x)) \quad (1.1)$$

where x represents the decision variables and $f_i(x)$ are the objective functions.

A solution is said to be **Pareto optimal** if no other solution improves one objective without degrading at least one other objective. The set of all such solutions forms the **Pareto front**, which represents the optimal trade-offs among objectives.

1.2 Motivation of the Work

The motivation of this work arises from the following challenges:

- Real-world problems involve multiple conflicting objectives
- Traditional optimization methods fail to balance these objectives effectively
- NSGA-III ensures diversity but suffers from slow convergence
- Efficient exploration and faster convergence are required

To address these challenges, we propose a hybrid approach that combines NSGA-III with Differential Evolution, leading to improved convergence speed and better-quality solutions.

1.3 Objective of the Work

The main objectives of this work are:

- To study multiobjective optimization techniques such as NSGA-II and NSGA-III
- To develop a hybrid DE-NSGA-III algorithm
- To apply the proposed method to hyperparameter optimization in deep learning
- To evaluate performance using metrics such as accuracy, F1-score, and convergence

Chapter 2

Non-dominated Sorting Genetic Algorithm II (NSGA-II)

2.1 Introduction

Multiobjective optimization problems require finding a set of optimal trade-off solutions rather than a single optimal solution. Classical optimization methods are often inadequate for such problems, especially when the objective functions are non-convex, discontinuous, or non-differentiable.

Evolutionary algorithms have emerged as powerful tools for solving multiobjective optimization problems due to their population-based nature. Among them, the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)**, proposed by Deb et al. (2002), is one of the most widely used algorithms.

NSGA-II efficiently approximates the Pareto front by combining three key ideas:

- Non-dominated sorting
- Crowding distance (diversity preservation)
- Elitism

2.2 Basic Concepts

2.2.1 Dominance

A solution x_1 is said to dominate another solution x_2 if:

- $f_i(x_1) \leq f_i(x_2)$ for all objectives i , and
- $f_j(x_1) < f_j(x_2)$ for at least one objective j .

2.2.2 Pareto Front

The set of non-dominated solutions forms the **Pareto front**, representing optimal trade-offs among objectives.

2.3 Non-dominated Sorting

NSGA-II classifies the population into different levels called **fronts**:

- F_1 : Non-dominated solutions (best front)
- F_2 : Solutions dominated only by those in F_1
- F_3 : Solutions dominated by F_1 and F_2 , and so on

Each solution is assigned a **rank** based on its front number.

2.4 Crowding Distance

To maintain diversity among solutions, NSGA-II uses a metric called **crowding distance**. For each solution:

- Sort solutions according to each objective
- Assign infinite distance to boundary solutions
- Compute distance based on neighboring solutions

A higher crowding distance indicates that the solution lies in a less crowded region of the Pareto front.

2.5 Selection Mechanism

NSGA-II uses **binary tournament selection** based on:

- Rank (priority to lower rank)
- Crowding distance (priority to higher distance)

2.6 Genetic Operators

2.6.1 Crossover

Crossover combines two parent solutions to generate offspring. It helps in exploring new regions of the search space.

2.6.2 Mutation

Mutation introduces random changes in solutions to maintain diversity and avoid premature convergence.

2.7 NSGA-II Algorithm

Input: Population size N , number of generations T

Output: Approximate Pareto front

1. Initialize population P_0 with N random solutions
2. Evaluate objective functions for all individuals
3. Perform non-dominated sorting to assign ranks
4. For each generation $t = 0, 1, 2, \dots, T$:

(a) Generate offspring population Q_t using:

- Selection
- Crossover
- Mutation

(b) Combine populations:

$$R_t = P_t \cup Q_t \tag{2.1}$$

(c) Perform non-dominated sorting on R_t to obtain fronts $F_1, F_2, \dots$

(d) Initialize new population $P_{t+1} = \emptyset$

(e) Add fronts sequentially:

- Add $F_1, F_2, \dots$ until population size exceeds N

(f) If last front exceeds limit:

- Sort solutions using crowding distance
 - Select top solutions to maintain size N
5. Return final non-dominated solutions from P_T

2.8 Advantages of NSGA-II

- Efficient non-dominated sorting mechanism
- Maintains diversity using crowding distance
- Elitism ensures best solutions are preserved
- Works well for problems with 2–3 objectives

2.9 Limitations of NSGA-II

Despite its effectiveness, NSGA-II has certain limitations:

- Performance degrades for many-objective problems (more than 3 objectives)
- Most solutions become non-dominated, reducing selection pressure
- Crowding distance becomes less effective in high-dimensional objective space
- Difficulty in maintaining well-distributed Pareto fronts

2.10 Motivation for NSGA-III

Due to the limitations of NSGA-II in handling many-objective optimization problems, an improved algorithm called **NSGA-III** was developed.

NSGA-III introduces reference-point-based selection to ensure better diversity and distribution of solutions across the Pareto front.

This motivates the next chapter, where NSGA-III is discussed in detail.

Chapter 3

Non-dominated Sorting Genetic Algorithm III (NSGA-III)

3.1 Introduction

As discussed in the previous chapter, NSGA-II is a powerful evolutionary algorithm for solving multiobjective optimization problems. However, its performance deteriorates when dealing with **many-objective problems** (i.e., problems with four or more objectives). In such cases, most solutions tend to become non-dominated, which reduces the effectiveness of non-dominated sorting and crowding distance mechanisms.

To overcome these limitations, Deb and Jain (2014) proposed the **Non-dominated Sorting Genetic Algorithm III (NSGA-III)**. This algorithm extends NSGA-II by introducing a **reference-point-based selection mechanism** to maintain diversity in high-dimensional objective spaces.

3.2 Motivation for NSGA-III

In many-objective optimization:

- Most solutions become non-dominated
- Selection pressure is reduced
- Crowding distance fails to maintain diversity

NSGA-III addresses these issues by:

- Introducing structured reference points
- Ensuring uniform distribution of solutions
- Maintaining diversity across the Pareto front

3.3 Key Idea of NSGA-III

The key idea of NSGA-III is to guide the selection process using a predefined set of **reference points** in the objective space.

- These reference points are uniformly distributed on a simplex.
- Each reference point represents a direction in objective space.
- Solutions are associated with the nearest reference point.

The algorithm ensures that each reference point is represented by at least one solution, leading to a well-distributed Pareto front.

3.4 Reference Points Generation

Reference points are generated using systematic methods such as the **Das and Dennis approach**. These points lie on a normalized hyperplane:

$$\sum_{i=1}^m f_i = 1 \quad (3.1)$$

where m is the number of objectives.

These points help guide the population toward diverse regions of the objective space.

3.5 Normalization of Objectives

Before associating solutions with reference points, objective values are normalized:

- Compute ideal point (minimum of each objective)
- Translate objective values
- Scale them to a common range

This ensures fair comparison among objectives with different scales.

3.6 Association with Reference Points

Each solution is associated with the nearest reference point based on perpendicular distance.

- Compute distance from each solution to all reference points
- Assign solution to closest reference point

3.7 Niching Mechanism

The niching process ensures diversity:

- Count number of solutions associated with each reference point (niche count)
- Prefer reference points with fewer associated solutions
- Select solutions from underrepresented regions

This helps maintain a uniform spread of solutions.

3.8 NSGA-III Algorithm

Input: Population size N , number of generations T

Output: Pareto-optimal solutions

1. Initialize population P_0 with N solutions
2. Generate reference points
3. Evaluate objective functions
4. For each generation $t = 0, 1, \dots, T$:
 - (a) Generate offspring Q_t using:
 - Selection
 - Crossover
 - Mutation
 - (b) Combine populations:

$$R_t = P_t \cup Q_t \tag{3.2}$$

(c) Perform non-dominated sorting:

$$R_t \rightarrow F_1, F_2, F_3, \dots$$

(d) Fill new population P_{t+1} :

- Add fronts sequentially until size exceeds N

(e) If last front exceeds capacity:

- Normalize objective values
- Associate solutions with reference points
- Perform niching:
 - Select solutions from least crowded reference points

5. Return non-dominated solutions

3.9 Advantages of NSGA-III over NSGA-II

NSGA-III provides several improvements over NSGA-II:

- **Better Diversity:** Uses reference points instead of crowding distance
- **Scalability:** Works well for many-objective problems (4+ objectives)
- **Uniform Distribution:** Ensures well-spread Pareto front
- **Effective Selection:** Maintains selection pressure even in high dimensions

3.10 NSGA-II vs NSGA-III

Feature	NSGA-II	NSGA-III
Diversity Mechanism	Crowding Distance	Reference Points
Suitable Objectives	2–3	4 or more
Selection Pressure	Weak in high dimension	Strong
Pareto Distribution	Uneven	Uniform

3.11 Summary

NSGA-III is a powerful extension of NSGA-II designed to handle many-objective optimization problems efficiently. By introducing reference-point-based selection and niching mechanisms, it ensures both convergence and diversity.

In this work, NSGA-III forms the backbone of the proposed hybrid algorithm, providing global exploration capabilities while other components such as proximal gradient and local search enhance local refinement.

Chapter 4

Differential Evolution based NSGA-III (DE-NSGA-III)

4.1 Introduction

While NSGA-III is an effective evolutionary algorithm for solving many-objective optimization problems, it relies on traditional genetic operators such as crossover and mutation. These operators provide good diversity but often lead to slow convergence, especially in high-dimensional search spaces.

To overcome this limitation, hybrid approaches have been developed by integrating alternative evolutionary strategies. One such approach is the incorporation of **Differential Evolution (DE)** into NSGA-III, resulting in the **DE-NSGA-III algorithm**.

DE-NSGA-III combines the strong exploration capability of NSGA-III with the fast convergence properties of Differential Evolution, leading to improved optimization performance.

4.2 Overview of Differential Evolution

Differential Evolution is a population-based optimization algorithm designed for continuous optimization problems. It generates new candidate solutions using vector differences between individuals.

The main steps in DE are:

- **Mutation**
- **Crossover**

- Selection

4.2.1 Mutation

For each target vector x_i , a mutant vector is generated as:

$$v_i = x_{r1} + F \cdot (x_{r2} - x_{r3}) \quad (4.1)$$

where:

- x_{r1}, x_{r2}, x_{r3} are randomly selected distinct individuals
- F is the scaling factor

4.2.2 Crossover

The trial vector u_i is generated by combining the target vector and mutant vector:

$$u_{i,j} = \begin{cases} v_{i,j}, & \text{if } rand_j \leq CR \\ x_{i,j}, & \text{otherwise} \end{cases} \quad (4.2)$$

where CR is the crossover probability.

4.2.3 Selection

The better solution between the trial vector and target vector is selected based on fitness.

4.3 Integration of DE with NSGA-III

In DE-NSGA-III, the traditional crossover and mutation operations of NSGA-III are replaced or augmented with DE-based operators.

- Population is initialized as in NSGA-III
- DE mutation and crossover are applied to generate offspring
- NSGA-III selection mechanism is used for survival

Thus, DE enhances exploration, while NSGA-III ensures diversity through reference points.

4.4 Working Principle of DE-NSGA-III

1. Initialize population P_0
2. Generate reference points
3. Evaluate objective functions
4. For each generation:
 - (a) Apply DE mutation
 - (b) Apply crossover
 - (c) Evaluate offspring
 - (d) Combine populations
 - (e) Perform non-dominated sorting
 - (f) Apply NSGA-III selection (niching)
5. Repeat until termination

4.5 Algorithm

Input: Population size N , generations T , scaling factor F , crossover rate CR

Output: Pareto-optimal solutions

1. Initialize population P
2. Generate reference points
3. For $t = 1$ to T :
 - (a) For each individual x_i :
 - Select x_{r1}, x_{r2}, x_{r3}
 - Mutation: $v_i = x_{r1} + F(x_{r2} - x_{r3})$
 - Crossover: $u_i = \text{crossover}(x_i, v_i)$
 - (b) Evaluate offspring
 - (c) Combine populations: $R = P \cup Q$

- (d) Perform non-dominated sorting
 - (e) Apply NSGA-III selection
 - (f) Update population
4. Return Pareto front

4.6 Advantages of DE-NSGA-III

- Faster convergence due to vector-based updates
- Better exploration of the search space
- Improved solution quality
- Maintains diversity using reference points

4.7 Comparison with NSGA-III

Feature	NSGA-III	DE-NSGA-III
Search Mechanism	Genetic Operators	Differential Evolution
Convergence Speed	Moderate	Faster
Exploration Ability	Good	Enhanced
Diversity	Reference Points	Reference Points

4.8 Pareto Front Comparison

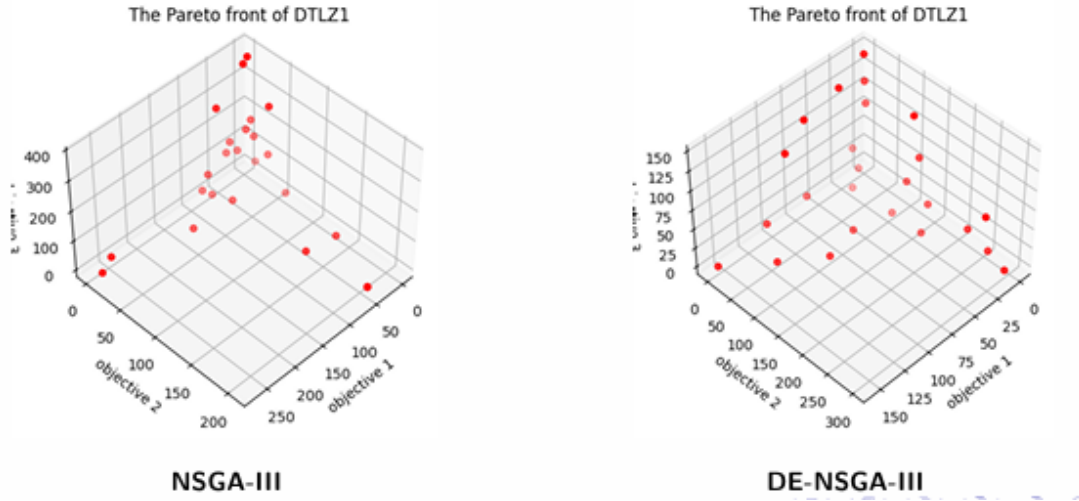


Figure 4.1: Comparison of Pareto fronts on DTLZ1 problem using NSGA-III and DE-NSGA-III

Observation:

- DE-NSGA-III produces a more uniform Pareto front
- Solutions are closer to the true Pareto optimal region
- NSGA-III shows less refined distribution
- DE improves both convergence and exploration

4.9 Application in This Work

DE-NSGA-III is applied to hyperparameter optimization of an MLP model on the MNIST dataset.

4.10 Summary

DE-NSGA-III improves convergence speed and solution quality while maintaining diversity, making it highly suitable for multiobjective optimization problems.

Chapter 5

Application: Multiobjective Hyperparameter Optimization

5.1 Introduction

To evaluate the effectiveness of the proposed hybrid optimization algorithm, we apply it to a real-world problem in deep learning, namely **multiobjective hyperparameter optimization**. As highlighted in the presentation, hyperparameter tuning is inherently a multiobjective problem, where multiple conflicting objectives such as accuracy, F1-score, and model complexity must be optimized simultaneously.

Traditional approaches such as grid search and random search are either computationally expensive or inefficient in exploring the search space. Moreover, single-objective optimization methods fail to capture the trade-offs among different performance metrics. Therefore, evolutionary multiobjective algorithms such as NSGA-III and its hybrid variant DE-NSGA-III are well-suited for this problem.

5.2 Problem Statement

The hyperparameter optimization problem can be formulated as:

$$\min / \max f_1(x), f_2(x), f_3(x), \quad x \in U \quad (5.1)$$

where:

- $f_1(x)$ = Accuracy (maximize)

- $f_2(x)$ = F1-score (maximize)
- $f_3(x)$ = Model complexity (minimize)

This formulation captures the trade-off between performance and efficiency, which is essential in real-world applications.

5.3 Why NSGA-III and DE-NSGA-III?

As discussed in the presentation:

- Grid search is computationally expensive
- Random search does not efficiently explore the search space
- Single-objective methods ignore trade-offs

NSGA-III provides:

- Diversity using reference points
- Well-distributed Pareto front
- Simultaneous optimization of multiple objectives

DE-NSGA-III further improves:

- Faster convergence using differential mutation
- Better exploration of the search space
- Improved solution quality

5.4 Dataset: MNIST

The MNIST dataset is used as a benchmark for evaluating the proposed approach.

- 70,000 grayscale images of handwritten digits (0–9)
- Image size: 28×28
- 60,000 training samples and 10,000 testing samples

The dataset is relatively simple compared to more complex datasets, making it suitable for benchmarking optimization algorithms.

5.5 DE-NSGA-III Framework

The overall framework integrates multiobjective optimization with deep learning:

- Generate initial population of hyperparameters
- Apply DE-NSGA-III for optimization
- Evaluate each solution using MLP model performance
- Select optimal solutions based on Pareto optimality

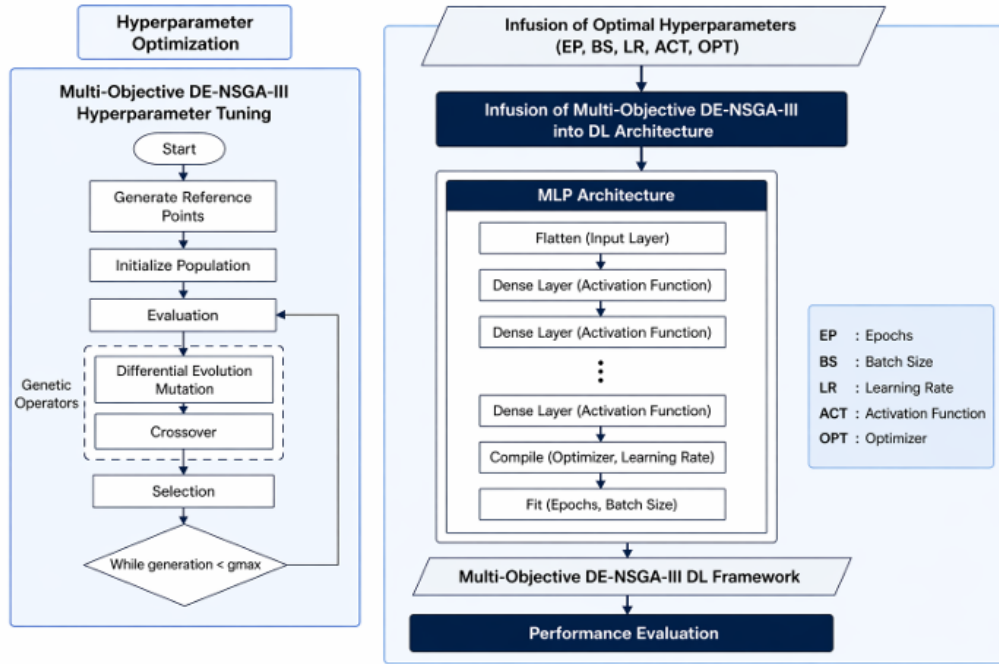


Figure 5.1: DE-NSGA-III Framework for Hyperparameter Optimization

5.6 Model Architecture: Multi-Layer Perceptron (MLP)

In this work, we use a **Multi-Layer Perceptron (MLP)** model.

5.6.1 Input Layer

Each MNIST image is flattened into:

$$28 \times 28 = 784 \text{ features} \quad (5.2)$$

5.6.2 Hidden Layers

- Fully connected dense layers
- Activation functions: ReLU or ELU

5.6.3 Output Layer

- 10 neurons corresponding to 10 digit classes
- Softmax activation for classification

5.7 Hyperparameters

The following hyperparameters are optimized:

- Learning rate
- Optimizer
- Batch size
- Number of epochs

These hyperparameters significantly influence model performance and training efficiency.

5.8 Summary

In this chapter, we formulated hyperparameter tuning as a multiobjective optimization problem and applied the DE-NSGA-III algorithm to solve it. The MNIST dataset and MLP model provide a suitable testbed for evaluating the effectiveness of the proposed hybrid optimization approach.

Chapter 6

Evaluation Metrics

6.1 Introduction

To evaluate the performance of the proposed DE-NSGA-III based hybrid optimization algorithm, several classification metrics are used. These metrics are derived from the confusion matrix and provide a comprehensive understanding of the model performance in terms of accuracy, precision, recall, and overall balance.

As discussed in the presentation, hyperparameter tuning in deep learning is formulated as a multiobjective optimization problem. Therefore, these metrics are also combined using a weighted formulation to evaluate the overall performance of solutions.

6.2 Confusion Matrix Basics

All evaluation metrics are derived from the confusion matrix, which consists of:

- True Positive (TP): Correctly classified instances
- True Negative (TN): Correctly rejected instances
- False Positive (FP): Incorrectly classified positive instances
- False Negative (FN): Incorrectly classified negative instances

6.3 Accuracy

Accuracy measures the overall correctness of the classification model.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6.1)$$

Interpretation:

- Represents the proportion of correctly classified samples
- Suitable for balanced datasets such as MNIST

6.4 Precision

Precision measures the proportion of predicted positive instances that are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6.2)$$

Interpretation:

- High precision indicates fewer false positives
- Important when incorrect positive predictions are costly

6.5 Recall

Recall measures the proportion of actual positive instances that are correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6.3)$$

Interpretation:

- High recall indicates fewer false negatives
- Important in scenarios where missing true cases is critical

6.6 F1-Score

F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6.4)$$

Interpretation:

- Provides a balance between precision and recall
- Particularly useful in multiobjective optimization

6.7 Model Complexity

In addition to classification performance, model complexity is also considered.

- Measured using number of model parameters
- Normalized for fair comparison across solutions

$$\text{NormParams} = \frac{\text{Number of Parameters}}{\text{Maximum Parameters}} \quad (6.5)$$

Lower values indicate simpler and more efficient models.

6.8 Weighted Sum Formulation

As presented in the PPT (Evaluation Metrics slide), the overall performance is computed using a weighted sum:

$$WS = w_1 \cdot F1 + w_2 \cdot \text{Accuracy} + w_3 \cdot \left(\frac{1}{\text{NormParams}} \right) \quad (6.6)$$

Explanation:

- $F1$ captures balance between precision and recall
- Accuracy measures overall classification performance
- $\frac{1}{\text{NormParams}}$ encourages simpler models

6.9 Choice of Weights

The selection of weights determines the importance of each objective.

6.9.1 Typical Values Used

Following the experimental setup:

$$w_1 = 0.4, \quad w_2 = 0.4, \quad w_3 = 0.2 \tag{6.7}$$

Justification:

- Equal importance to accuracy and F1-score
- Lower weight for complexity to avoid sacrificing performance

6.9.2 Alternative Weight Settings

- Accuracy-focused: (0.3, 0.5, 0.2)
- Balanced: (0.4, 0.4, 0.2)
- Complexity-sensitive: (0.3, 0.3, 0.4)

6.10 Role in Multiobjective Optimization

These metrics define the objective space for DE-NSGA-III optimization. The algorithm aims to:

- Maximize Accuracy
- Maximize F1-score
- Minimize model complexity

By optimizing these objectives simultaneously, the algorithm produces a set of Pareto-optimal solutions representing different trade-offs.

6.11 Summary

In this chapter, we presented the evaluation metrics used to assess the performance of the proposed DE-NSGA-III algorithm. These metrics, derived from the confusion matrix, provide a comprehensive evaluation of classification performance. The weighted sum formulation allows effective comparison of solutions in a multiobjective setting, ensuring a balance between accuracy, robustness, and model complexity.

Chapter 7

Results and Discussion

7.1 Introduction

In this chapter, we present the experimental results obtained using the proposed **DE-NSGA-III** algorithm. The performance of DE-NSGA-III is compared with the standard NSGA-III algorithm on the **MNIST dataset** using a Multi-Layer Perceptron (MLP) model. The evaluation is based on accuracy, F1-score, model complexity, convergence speed, and weighted sum.

7.2 Quantitative Results

Method	Accuracy	F1 Score	Parameters	Norm Params	Convergence	Weighted Sum
NSGA-III	0.9815	0.9814	101,770	1.0	8	2.9654
DE-NSGA-III	0.9829	0.9829	101,770	1.0	6	2.9658

Table 7.1: Performance comparison between NSGA-III and DE-NSGA-III on MNIST dataset

7.3 Analysis of Results

7.3.1 Accuracy Improvement

The DE-NSGA-III algorithm achieves an accuracy of 0.9829 compared to 0.9815 for NSGA-III.

- This indicates a measurable improvement in classification performance.
- The improvement is due to the efficient exploration capability of Differential Evolution.

7.3.2 F1-Score Enhancement

The F1-score improves from 0.9814 to 0.9829.

- Indicates better balance between precision and recall
- Demonstrates improved classification robustness

7.3.3 Convergence Speed

- NSGA-III converges in 8 generations
- DE-NSGA-III converges in 6 generations

Observation:

- Faster convergence is achieved due to DE-based mutation
- Vector-based updates accelerate optimization

7.3.4 Model Complexity

- Both methods use the same number of parameters (101,770)
- Normalized parameter value remains constant

Interpretation:

- Performance improvement is achieved without increasing model complexity

7.3.5 Weighted Sum Comparison

- NSGA-III: 2.9654
- DE-NSGA-III: 2.9658

Observation:

- Higher weighted sum indicates better overall performance
- DE-NSGA-III provides a better trade-off among objectives

7.4 Pareto Front Analysis

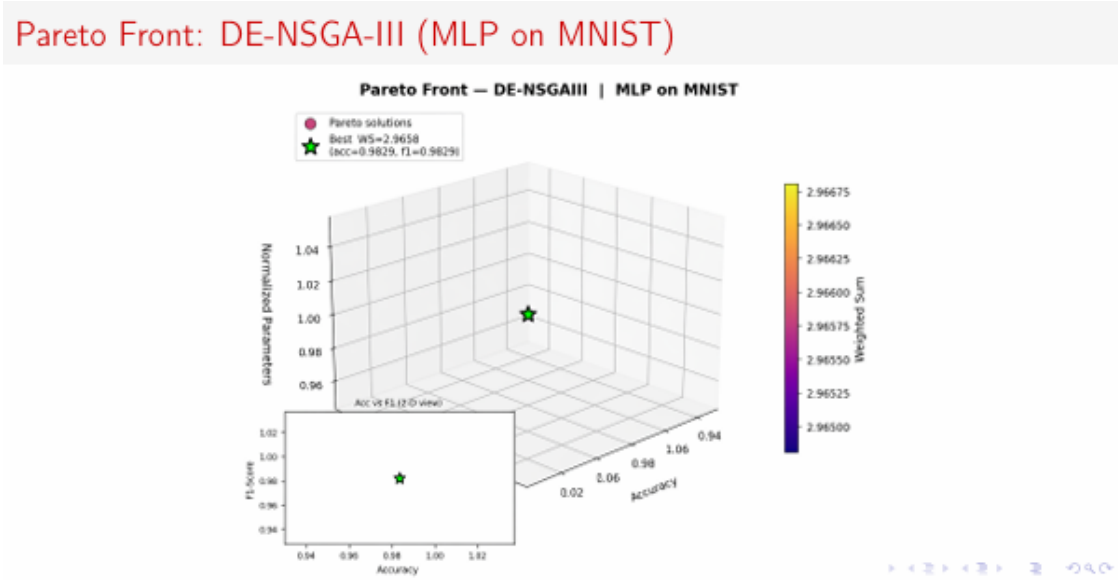


Figure 7.1: Pareto Front obtained using DE-NSGA-III on MNIST dataset

7.4.1 Observations from Pareto Front

- The Pareto front is concentrated near high accuracy and F1-score values
- The best solution (highlighted) achieves near-optimal performance
- Solutions show minimal variation in complexity due to normalization

7.4.2 Comparison with NSGA-III

- NSGA-III provides good diversity but slower convergence
- DE-NSGA-III shifts solutions toward better optimal regions
- Improved convergence and solution quality are observed

7.5 Key Observations

- DE-NSGA-III achieves better accuracy and F1-score
- Faster convergence (6 vs 8 generations)

- Improved weighted sum performance
- No increase in model complexity
- Better optimization efficiency

7.6 Discussion

The results demonstrate that integrating Differential Evolution with NSGA-III significantly improves optimization performance.

- NSGA-III ensures diversity through reference points
- Differential Evolution enhances convergence speed
- The hybrid approach improves both exploration and exploitation

7.7 Summary

The experimental results confirm that DE-NSGA-III outperforms standard NSGA-III in terms of accuracy, convergence speed, and overall performance. This validates the effectiveness of the proposed hybrid approach for multiobjective optimization in deep learning.

Chapter 8

Conclusion

8.1 Summary of the Work

In this thesis, we studied the problem of **multiobjective optimization** in the context of deep learning, with a focus on hyperparameter tuning of neural network models. Hyperparameter tuning is inherently a multiobjective problem, as it requires balancing multiple conflicting objectives such as accuracy, F1-score, and model complexity.

Traditional optimization methods such as grid search and random search are inefficient and fail to capture the trade-offs among these objectives. To address this limitation, evolutionary algorithms such as NSGA-II and NSGA-III have been widely used.

NSGA-III, in particular, provides improved diversity and well-distributed Pareto-optimal solutions using reference-point-based selection. However, it suffers from relatively slow convergence due to the use of traditional genetic operators.

8.2 Proposed Approach

To overcome this limitation, we proposed a hybrid algorithm, namely **DE-NSGA-III**, which integrates Differential Evolution (DE) with NSGA-III.

- NSGA-III ensures diversity using reference points
- Differential Evolution improves convergence speed
- The hybrid approach enhances exploration of the search space

This combination results in an efficient multiobjective optimization framework.

8.3 Experimental Findings

The proposed DE-NSGA-III algorithm was evaluated on the **MNIST dataset** using a Multi-Layer Perceptron (MLP) model.

The experimental results show:

- Accuracy improved from 0.9815 to 0.9829
- F1-score improved from 0.9814 to 0.9829
- Faster convergence (6 generations compared to 8)
- Improved overall performance based on weighted sum

These results demonstrate that DE-NSGA-III achieves better performance without increasing model complexity.

8.4 Key Insights

The key findings of this work are:

- Hyperparameter tuning in deep learning is a multiobjective optimization problem
- NSGA-III effectively balances accuracy, F1-score, and model complexity
- Differential Evolution significantly improves convergence speed
- DE-NSGA-III provides better exploration and solution quality
- Hybrid optimization produces better trade-offs than standalone methods

8.5 Conclusion

The results confirm that **DE-NSGA-III outperforms standard NSGA-III** in terms of accuracy, convergence speed, and overall performance. By combining the strengths of evolutionary algorithms and Differential Evolution, the proposed method achieves a better balance between exploration and exploitation.

This makes DE-NSGA-III a powerful and efficient approach for solving complex multiobjective optimization problems in deep learning.

8.6 Future Work

There are several possible directions for future research:

- Extending the approach to more complex models such as CNNs and Transformers
- Applying the method to larger datasets such as CIFAR-100 and ImageNet
- Incorporating additional objectives such as training time and energy efficiency
- Developing adaptive parameter control strategies for DE

8.7 Final Remark

In conclusion, this work demonstrates that hybrid evolutionary optimization methods, particularly DE-NSGA-III, provide an effective and scalable solution for multiobjective optimization problems in modern machine learning applications.

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